# Glossary of Abbreviations for S+SSPR 2026 Paper

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| Abbrev. | Full Term | Description |
|---|---|---|
| **PCA** | **Principal Component Analysis** | A linear dimensionality-reduction technique that projects correlated variables onto orthogonal axes (principal components) ordered by explained variance. It works by eigendecomposing the covariance matrix of the standardized data; here it compresses 7 raw airline variables into 3–4 components while retaining 87.8–95.6% of total variance. |
| **pc** | **Principal Component** | A single output axis of a PCA decomposition (pc1, pc2, pc3, pc4), ranked by the share of variance it explains. Each is a linear combination of the original 7 variables, with "loadings" indicating how strongly each original variable contributes; the paper interprets pc1–pc4 substantively (growth, profit–wage tension, fuel shocks, COVID yield collapse). |
| **LF** | **Load Factor** | An airline-industry KPI equal to revenue passenger miles divided by available seat miles, expressed as a percentage — how full planes are on average. One of the seven raw clustering inputs; collapses from ∼85% to ∼59% in 2020, driving the COVID outlier cluster. |
| **PR** | **Participation Ratio** | A statistic (Eq. 1 in the paper) computed as $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, where $\lambda_i$ are the covariance-matrix eigenvalues. It estimates how many dimensions are "effectively" independent versus redundant; borrowed from physics (localization of quantum states) and used in ML as a collinearity/effective-dimensionality diagnostic. Here PR $\approx$ 2.46 of 7 nominal variables. |
| **Sil** | **Silhouette Coefficient** | For each point, $(b-a)/\max(a,b)$, where $a$ is mean intra-cluster distance and $b$ is mean distance to the nearest other cluster; averaged over all points for the score. Ranges $-1$ to 1, higher is better. Introduced by Rousseeuw (1987), cited as ref. [6]. |
| **DB** | **Davies–Bouldin Index** | For each cluster, the maximum ratio of (within-cluster scatter of cluster $i$ + within-cluster scatter of cluster $j$) to (distance between centroids $i$ and $j$) over all other clusters $j$, then averaged. Lower values indicate tighter, better-separated clusters. Introduced by Davies & Bouldin (1979), cited as ref. [1]. |
| **ARI** | **Adjusted Rand Index** | A chance-corrected version of the Rand Index, which counts pairs of points agreeing (both together or both separate) across two partitions. ARI adjusts for chance agreement, so 1.0 = identical partitions and $\approx$0 = random agreement. Introduced by Hubert & Arabie (1985), cited as ref. [3]. |
| **kpca / kPCA** | **Kernel PCA** | A nonlinear generalization of PCA: instead of eigendecomposing the covariance matrix directly, it eigendecomposes a kernel (Gram) matrix of pairwise similarities computed via a kernel function (linear, RBF, polynomial, etc.), implicitly projecting into a higher-dimensional feature space where linear PCA is then applied. Introduced by Schölkopf, Smola & Müller (1998), cited as ref. [8]. |
| **RBF** | **Radial Basis Function (kernel)** | A kernel of the form $K(x,y) = \exp(-\gamma \lVert x-y \rVert^2)$, producing similarity that decays smoothly and symmetrically with Euclidean distance. Widely used as a default nonlinear kernel in kernel PCA, SVMs, and kernel ridge regression because it can approximate arbitrarily smooth functions. |

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| $\gamma$ (gamma) | **Kernel bandwidth** | The scale parameter in the RBF kernel controlling how quickly similarity decays with distance; small $\gamma$ = smoother/wider similarity, large $\gamma$ = sharper/narrower. Set here via the median pairwise-distance heuristic (a common data-driven default avoiding cross-validation). |
| **5-NN** | **5-Nearest Neighbors** | A graph-construction rule where each point is connected only to its 5 closest points by distance, producing a sparse local-connectivity graph rather than a fully dense similarity matrix. Used as the basis for the graph-diffusion kernel, where similarity propagates only along these local edges. |
| **FDA** | **Fisher Discriminant Analysis** | A supervised technique that finds the linear projection maximizing the ratio of between-class variance to within-class variance, given known class labels. The paper argues it's unsuitable here since it requires pre-labelled clusters (circular for structure-discovery) and is numerically unstable with only ∼5–9 points per cluster. |
| **LDA** | **Linear Discriminant Analysis** | A supervised classification and dimensionality-reduction method closely related to FDA, projecting data to maximize class separability. Mentioned as a possible future validation step once labelled data is available, to check whether supervised projections align with the unsupervised PCA findings. |
| **LOOCV** | **Leave-One-Out Cross-Validation** | A cross-validation scheme where, for $n$ observations, the model is trained $n$ times, each time leaving out exactly one point for testing and training on the rest — an extreme case of $k$-fold CV with $k = n$. Used with a nested grid search in the kernel ridge regression check. |
| $\mathbf{R^2}$ | **Coefficient of Determination** | The proportion of variance in the dependent variable (here, Operating Profit) explained by the model's predictions, ranging up to 1.0 for a perfect fit (can go negative for a poor model). Reported per model in Table 5, both including and excluding the 2020 outlier year. |
| **HMM** | **Hidden Markov Model** | A statistical model of sequential data assuming an unobserved ("hidden") state process that evolves via Markov transitions, with each state emitting observable outputs. Proposed as future work to quantify regime persistence and transition probabilities between the paper's profit-cycle clusters. |
| **VAR** | **Vector Autoregression** | A multivariate time-series model where each variable is regressed on lagged values of itself and every other variable in the system, up to lag order $p$. Proposed as a "regime-switching VAR," where coefficients depend on a latent regime state (linking to the HMM idea). Would need far more than the current $n = 26$ yearly observations given the parameter count scales with $n_{\text{vars}}^2 \times$ lag order. |
| **t-SNE** | **t-Distributed Stochastic Neighbor Embedding** | Converts high-dimensional pairwise distances into neighbor probabilities via a Gaussian kernel (tuned by a **perplexity** parameter setting effective neighborhood size), then fits a low-dimensional layout whose neighbor probabilities — computed with a heavy-tailed Student's t-distribution — best match, minimizing KL divergence. The heavy tails avoid the "crowding problem" of squeezing high-D neighborhoods into 2D. Purely local/qualitative: axes and inter-cluster distances aren't meaningfully interpretable. |
| **UMAP** | **Uniform Manifold Approximation and Projection** | Builds a weighted $k$-nearest-neighbor graph approximating the data's local manifold topology (with an adjustable **n_neighbors** parameter), then optimizes a low-dimensional embedding via cross-entropy loss to reproduce that fuzzy topological structure. Generally faster than t-SNE and often considered better at preserving some global structure, though results remain stochastic and axes uninterpretable. |